Md Rezwan Bin Mizan

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Academic Appointments

Postdoctoral Research Associate *Oct 2026 – Present*
Department of Mathematics, Virginia Tech

- Supervisor: Prof. Traian Iliescu

Research Assistant *Jun 2024 – Aug 2026*
Department of Mathematics, University of Houston

Instructor of Record *Aug 2025 – May 2026*
Department of Mathematics, University of Houston

Graduate Teaching Assistant *Aug 2021 – May 2024*
Department of Mathematics, University of Houston

Lecturer *Jan 2018 – Jul 2021*
American International University–Bangladesh (AIUB)

Education

Ph.D. in Mathematics *Fall 2021 – Summer 2026*
University of Houston, Texas, USA

- Dissertation: *Parametric Low-Rank Model Reduction for Nonlinear Fluid Systems*
- Advisor: Prof. Maxim Olshanskii Co-Advisor: Prof. Ilya Timofeyev
- Defended June 16, 2026 GPA: 3.746/4.00

M.S. in Applied Mathematics *Dec 2025*
University of Houston, Texas, USA

- GPA: 3.73/4.00

M.S. in Applied Mathematics *2016 – 2017*
University of Dhaka, Bangladesh

- Thesis: Spectral Relaxation Approach on Convective Heat Transfer of Magnetic and Non-Magnetic Particles over a Stretching Cylinder with Induced Magnetic Enhancement
- GPA: 3.90/4.00

B.S. in Mathematics *2010 – 2015*
University of Dhaka, Bangladesh

- Project: A Study of Differential Equations in Population Biology
- CGPA: 3.70/4.00

Research Interests

- Reduced-order modeling for nonlinear and parametric partial differential equations
- Tensor-based methods and machine-learning-enhanced ROMs (DL-ROM, hybrid approaches)
- Computational fluid dynamics, including non-Newtonian and viscoplastic (Herschel–Bulkley) flows
- Large-scale scientific computing and efficient simulation of complex systems

Publications

Published

- **Reduced-Order Modeling of Parameterized Visco-Plastic Shallow Flows.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  International Journal for Numerical Methods in Fluids, 2026.
  DOI: 10.1002/fld.70105
- **Comparison of a Parametric Physics-Informed Neural Network and a Tensorial Reduced-Order Model for the Shallow-Water Dam-Break Problem.**
  Anton Myshak, **Md Rezwan Bin Mizan**, and Ilya Timofeyev.
  Fluids, 2026.
  DOI: 10.3390/fluids11100251
- **A Parametric Tensor ROM for the Shallow Water Dam Break Problem.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  Computers & Fluids, 2026.
  DOI: 10.1016/j.compfluid.2026.107005
- **Parametric Reduced Order Models for the Generalized Kuramoto–Sivashinsky Equations.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  Applied Mathematics and Computation, 2026.
  DOI: 10.1016/j.amc.2026.130196
- **Mathematical and Buongiorno Nanofluid Modelling to Predict the Thermal and Solutal Performance of Magneto-Bioconvective Casson Nanofluid Flowing in a Rotatory Disk.**
  Sulyman O. Salawu, MD Shamshuddin, **Md Rezwan Bin Mizan**, and Sajjadul Bari.
  Physica Scripta, 2024.
  DOI: 10.1088/1402-4896/ad8e1a
- **Computation of Ferromagnetic/Nonmagnetic Nanofluid Flow Over a Stretching Cylinder with Induction and Curvature Effects.**
  Md Rezwan Bin Mizan, Mohammad Ferdows, MD Shamshuddin, et al.
  Heat Transfer, 2021.
  DOI: 10.1002/htj.22122

Under Review

- **Tensorial Reduced-Order Models for Parametric Coupled Reaction-Diffusion Systems: Application to Brain Tumor Growth Modeling.**
  Asikul Islam, **Md Rezwan Bin Mizan**, Maxim Olshanskii, and Andreas Mang.
  Computers and Mathematics with Applications, under review, 2026.
  Preprint: arXiv:2603.14101

Dissertation

- **Parametric Low-Rank Model Reduction for Nonlinear Fluid Systems.**
  Ph.D. Dissertation, University of Houston, 2026.
  UH Institutional Repository

Teaching Experience

Instructor of Record *Aug 2025 – May 2026*
University of Houston

- Designed and delivered a computation-driven linear algebra course that integrates theory with hands-on MATLAB applications.
- Developed assignments, exams, and coding projects that build conceptual understanding and problem-solving skills.
- Mentored students through office hours and supported diverse learning needs.
  Fall 2025, Spring 2026: MATH 2131 – Linear Algebra with MATLAB

Graduate Teaching Assistant *Aug 2021 – May 2024*
University of Houston

- Led Calculus I–III lab sessions and supported instruction across multiple undergraduate courses.
- Provided tutoring at CASA and assisted students with diverse mathematical backgrounds.
- Evaluated coursework in calculus, discrete mathematics, probability, and linear algebra.

Fall 2021:	MATH 2413 – Calculus I
Spring 2022:	MATH 2413 – Calculus I; MATH 3336 – Discrete Mathematics
Summer 2022:	MATH 3338 – Probability *(grader/tutor)*
Fall 2022:	MATH 3338 – Probability *(grader/tutor)*
Spring 2023:	MATH 2414 – Calculus II
Summer 2023:	MATH 4380 – Survey of Mathematics *(grader/tutor)*
Fall 2023:	MATH 4380 – Survey of Mathematics *(grader/tutor)*
Spring 2024:	MATH 2305 – Discrete Mathematics
Fall 2024:	MATH 2318 – Linear Algebra
Summer 2025:	MATH 2415 – Calculus III

Lecturer *Jan 2018 – Jul 2021*
American International University–Bangladesh (AIUB)

- Taught undergraduate courses including Calculus, Differential Equations, Numerical Methods, and Complex Variables.
- Designed course materials and assessments aligned with learning objectives.
- Transitioned courses to online delivery during COVID-19 and mentored students on academic progress.

Conference and Seminar Talks

- **A Parametric Tensor ROM for the Shallow Water Dam Break Problem**
  Minisymposium, SIAM TX–LA Conference, University of Texas at Austin, Sep 2025.
- **Tensorial Reduced Order Modeling of Viscoplastic Shallow-Water Dam-Break Flows**
  Scientific Computing Seminar, University of Houston, Sep 2025.
- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation**
  Graduate Student Paper Presentation, University of Houston, Apr 2024.

Poster Presentations

- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation**
  SIAM TX–LA Section Meeting, Baylor University, Oct 2024.
- **Reduced Order Modeling of Kuramoto–Sivashinsky Equation using Proper Orthogonal Decomposition**
  Mathematical Opportunities in Digital Twins (Math-DT), George Mason University, Dec 2023.

Honors & Fellowships

- NSF Research Assistantship (DMS-2309197, DMS-2408978), University of Houston *2024 – 2026*
- Graduate Tuition Fellowship, University of Houston *2021 – 2026*
- Teaching Assistantship, University of Houston *2021 – 2024*
- Bangladesh Sweden Trust Fund Award, Ministry of Finance, Bangladesh *2021*
- National Science and Technology Fellowship, Government of Bangladesh *2016 – 2017*

Leadership & Service

Treasurer *2023 – 2026*
AMS Student Chapter, University of Houston

- Managed chapter finances and supported organization of academic and student-focused events.

Public Relations & Publicity Secretary *2025 – 2026*
Bangladesh Graduate Student Association, University of Houston

- Coordinated communications and promoted events to enhance student engagement and outreach.

Professional Training

- CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning, University of Houston, 2025
- TAMIDS Scientific Machine Learning (SciML) Summer School, Texas A&M University, 2025
- Workshop on Presentation, Goal Setting, and Professional Development, Life Carnival Foundation
- Faculty Development Webinar on Outcome-Based Education (OBE), AIUB

Skills & Languages

- **Programming:** Python, MATLAB, Fortran, Mathematica
- **Libraries & Tools:** NumPy, SciPy, pandas, TensorLy, scikit-learn, PyTorch, Google Colab
- **Version Control:** Git, GitHub
- **Languages:** English (fluent), Bengali (native)

Memberships

- American Mathematical Society (AMS), Member *2022 – Present*
- Society for Industrial and Applied Mathematics (SIAM), Member *2023 – Present*